

Week 2 Practice: Python Warm-ups

AI Frameworks

Practice goals (80 minutes)

By the end of this session you should have:

1. Completed short exercises on lists and strings
2. Written functions for **mean**, **min**, and **max** (no pandas)
3. (Optional) Read a tiny CSV with the `csv` module
4. A clear note: “next week pandas will make tables easier”

Agenda

1. Warm-up drills (20–25 min)
2. Mean / min / max functions (25–30 min)
3. Optional: tiny CSV with `csv` (15 min)
4. Quiz + done checklist (10–15 min)

Drill 1 — lists of numbers

```
1 nums = [4, 1, 9, 2, 6]
2
3 # TODO: print the length, the first item, the last item
4 # TODO: append 0 and print the new list
5 # TODO: create a new list with only values > 5 (use a loop)
```

Check results by hand before asking for help.

Drill 2 — strings as sequences

```
1 word = "Frameworks"
2
3 print(len(word))
4 print(word[0], word[-1])
5
6 # TODO: count how many letters are lowercase 'a' (use a loop)
7 # TODO: build a new string with only the first 4 characters (slice)
```

Drill 3 — a list of dicts (mini “table”)

```
1 rows = [  
2     {"name": "Ada", "score": 90},  
3     {"name": "Alan", "score": 80},  
4     {"name": "Grace", "score": 95},  
5 ]  
6  
7 # TODO: print each name  
8 # TODO: compute the average score with a loop
```

This is the awkward version of a table—pandas will feel nicer next week.

Task — write three functions

Without using Python's `min`, `max`, or statistics libraries:

- `my_mean(values)`
- `my_min(values)`
- `my_max(values)`

Assume `values` is a non-empty list of numbers.

Starter template

```
1 def my_mean(values):
2     total = 0
3     for v in values:
4         total = total + v
5     return total / len(values)
6
7 def my_min(values):
8     pass # TODO
9
10 def my_max(values):
11     pass # TODO
12
13 data = [5, 2, 9, 2, 7]
14 print(my_mean(data), my_min(data), my_max(data))
```

Check yourself

- For [5, 2, 9, 2, 7]: mean should be 5.0; min 2; max 9
- Try a second list to be sure
- Optional stretch: what should happen on an empty list? (discuss—don't crash silently)

Try this

After yours work, compare one result to Python's built-in `min/max`.

Optional — peek at CSV

Create a file `tiny.csv` (or use one provided): `name,score`

`Ada,90`

`Alan,80`

`Grace,95`

Read with the csv module

```
1 import csv
2
3 scores = []
4 with open("tiny.csv", "r", newline="") as f:
5     reader = csv.DictReader(f)
6     for row in reader:
7         scores.append(float(row["score"]))
8
9 print(scores)
10 print(my_mean(scores))
```

Next week: `pd.read_csv("tiny.csv")` — one clear line.

If the optional part is slow

- Skip CSV and finish mean/min/max solidly
- Or use Colab's file upload / a path the instructor shares

Common mistake

Spending all remaining time fighting file paths. Core goal today = functions on lists.

Quiz — Practice check

```
1 def my_max(values):  
2     best = values[0]  
3     for v in values:  
4         if v > best:  
5             best = v  
6     return best  
7  
8 print(my_max([3, 8, 2]))
```

Quiz

What is printed?

1. 3
2. 8
3. 2
4. [3, 8, 2]

Answer

2. 8 — the loop keeps the largest value seen so far.

Done checklist

1. Warm-up drills on lists / strings attempted in a notebook
2. `my\_mean`, `my\_min`, `my\_max` work on a sample list
3. (Optional) CSV scores loaded and averaged
4. You can explain list vs dict in one sentence each

Before you leave

- Save `week02.ipynb`
- Keep your functions—handy to compare with pandas later

Next week

Lecture: pandas DataFrames

Practice: explore a friendly CSV with `head`, filters, simple questions